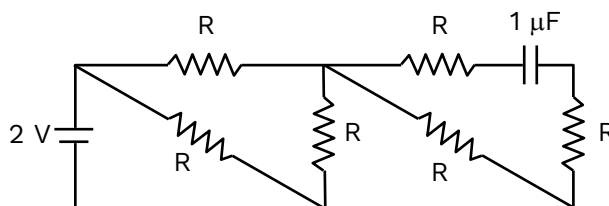
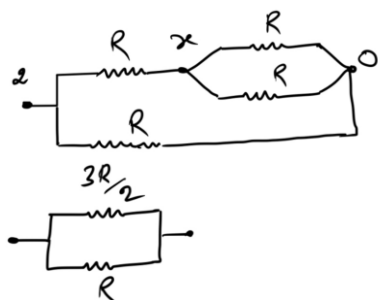


PHYSICS

Q1 Find current in the circuit at steady state, given $R = 2\Omega$.



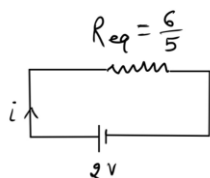
Sol.



$$\frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{\frac{3R}{2} \times R}{\frac{3R}{2} + R} = \frac{3R}{5}$$

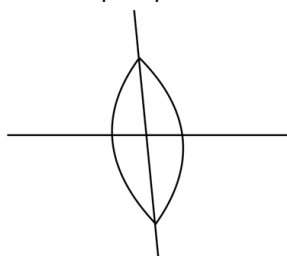
$$= \frac{3}{5}(2) = \frac{6}{5}$$



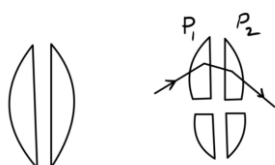
$$\Delta V = iR$$

$$2 = i \left(\frac{6}{5} \right) \Rightarrow i = \frac{10}{6} = \frac{5}{3}$$

Q2 A convex lens of power P is cut into four equal parts. Find the power of one part.



Sol.



$$P = P_1 + P_2$$

$$P = 2P_1$$

$$P_1 = \frac{P}{2}$$

Q3 A force $\vec{F} = 2\hat{i} + \hat{j} + 2\hat{k}$ acting at a point $(1, 1, 1)$. Find torque of this force about the origin.

Sol. $\vec{r} \times \vec{F}$

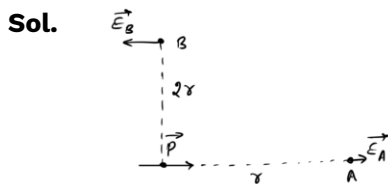
$$(\hat{i} + \hat{j} + \hat{k}) \times (2\hat{i} + \hat{j} + 2\hat{k})$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

$$= \hat{i}(1) - \hat{j}(2 - 2) + \hat{k}(1 - 2)$$

$$= \hat{i} - \hat{k}$$

Q4 The point A is situated on the axis of dipole at a distance 'r' from the dipole with E_0 and V_0 the electric field and electric potential at A. Find the electric field and potential at point B at distance '2r' from dipole on its perpendicular bisector in terms of E_0 and V_0 .



$$E_0 = \frac{2kP}{r^3}$$

$$V_0 = \frac{kP}{r^2}$$

$$E = \frac{kP}{(2r)^3}, V_B = 0$$

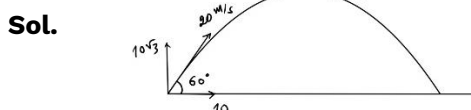
$$E_0 = \frac{2kP}{r^3}$$

$$E = \frac{kP}{8r^3}$$

$$\frac{E_0}{E} = 16$$

$$E = \frac{E_0}{16}$$

Q5 A projectile is thrown with a velocity 20 m/s at an angle of projection 60° from ground. Find Kinetic energy difference between point of projection and highest point, if mass of the particle is m.



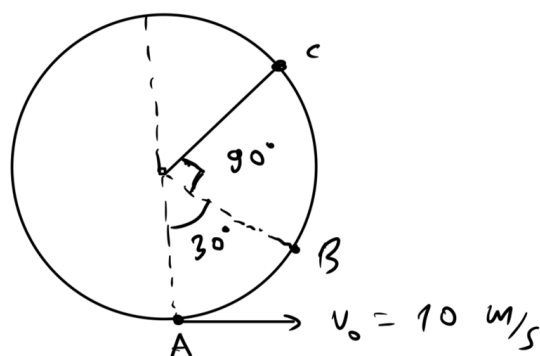
$$KE_A = \frac{1}{2}m(10)^2 + \frac{1}{2}m(10\sqrt{3})^2$$

$$KE_B = \frac{1}{2}m(10)^2 + 0$$

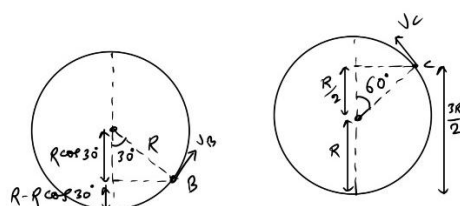
$$KE_A - KE_B = \frac{1}{2}m(300)$$

$$= 150 \text{ m}$$

Q6 Find the ratio of kinetic energy at point B and point C.



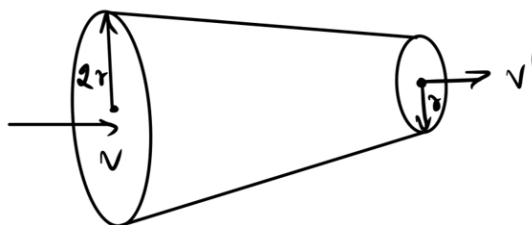
Sol.



$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}m(10)^2 + 0 = \frac{1}{2}mv^2 + mgh$$

Q7 A tube with non-uniform cross section has water inflow velocity 'v'. Calculate the velocity of water outflow v'.

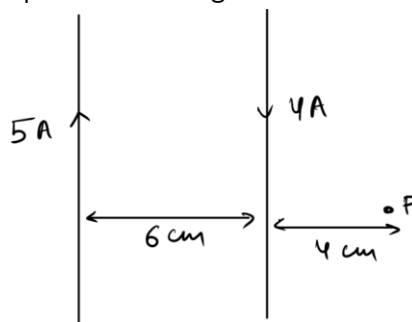


Sol. $A_1v_1 = A_2v_2$

$$\pi(2r)^2v = \pi r^2v'$$

$$v' = 4v$$

Q8 Find the net magnitude field at point P due to given infinite current carrying wires.



Sol. $B_{\text{net}} = B_1 - B_2$

$$B_1 = \frac{\mu_0 i_1}{2\pi d_1}, B_2 = \frac{\mu_0 i_2}{2\pi d_2}$$

$$I_1=4A, d_1 = 4\text{cm}, I_2=5A, d_2 = 10\text{cm}$$

$$\therefore B_{\text{net}} = 10 \mu\text{T}$$

Q9 A surface is incident with a photon of wavelength λ then kinetic energy of electron is 2 eV, if wavelength of photon is $\lambda/2$. Then maximum kinetic energy of ejected electron is: (Given work function of metal surface = 1 eV).

Sol. $KE_{\text{max}} = \frac{hc}{\lambda} - \phi$

Q10 Find the correct dimensional formula for the capacitance in terms of M, L, T and C where they stand for unit of mass, length, time and charge respectively.

Sol. $C = \frac{Q}{V} = \frac{Q}{\frac{Q}{F R}}$

$$= \frac{Q^2}{F R} = \frac{C^2}{M L T^{-2} \times L}$$

$$= \frac{Q^2}{F R} = \frac{C^2}{M L^2 T^{-2}}$$

$$V = \frac{kQ}{R}$$

$$F = \frac{kQ_1 Q_2}{R.R}$$

$$V = \frac{kQ}{R} = \frac{F R}{Q}$$

Q11 The displacement of a particle moving under the action of a force $\vec{F} = 2\hat{i} + b\hat{j} + \hat{k}$ is $\vec{d} = \hat{i} + \hat{j} + \hat{k}$. Find the value of b if the work done by the force is zero.

- (1) 0
(2) +3
(3) -3
(4) -1

Sol. $\vec{F} \cdot \vec{d} = 0$
 $2 + b + 1 = 0$
 $b = -3$

Q12 A proton is moving with uniform velocity of 2×10^8 m/s in uniform magnetic and electric fields which are perpendicular to each other. If electric field is switched off then proton moves in circular path of radius 1.6×10^{-5} m. The magnetic field B is:

- (1) $5 \times 10^{-5}\text{T}$
(2) $1.2 \times 10^5\text{T}$
(3) $2.5 \times 10^4\text{T}$
(4) $2.5 \times 10^2\text{T}$

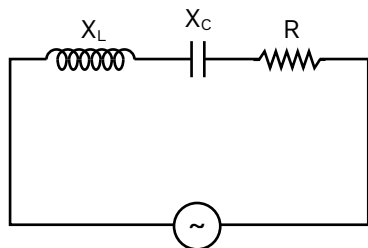
Sol. $R = \frac{mv}{qB}$

$$B = \frac{1.67 \times 10^{-27} \times 2 \times 10^8}{1.6 \times 10^{-19} \times 1.6 \times 10^{-5}}$$

$$= \frac{2}{1.6} \times 10^5 = \frac{5}{4} \times 10^5 = 1.25 \times 10^5$$

Q13 The peak current of a LCR circuit is I_0 . If the resistance of the circuit is doubled then what will be the value of peak current.

Sol.



$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$I_0 = \frac{V_0}{R}$$

$$I = \frac{V_0}{2R} = \frac{I_0}{2}$$

Q14 Assertion: In young's double slits experiment red fringe are wider compared to blue one.

Reason: Fringe width is directly proportional to wavelength.

(1) A is true but R is false.

(2) Both A and R are true but R is not the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A.

(4) A is false but R is true.

Sol. $\beta = \frac{\lambda D}{d}$

Q15 The maximum percentage error in the measurement of density of a wire is:

Given,

mass (m) = $(0.60 \pm 0.03)\text{g}$

Radius (r) = $(0.50 \pm 0.01)\text{cm}$

Length (l) = $(10.00 \pm 0.05)\text{cm}$

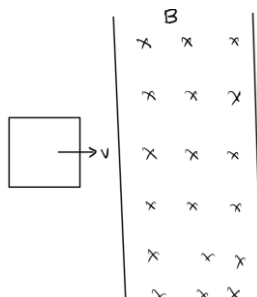
Sol. $\rho = \frac{m}{\pi r^2 L} \Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta L}{L}$

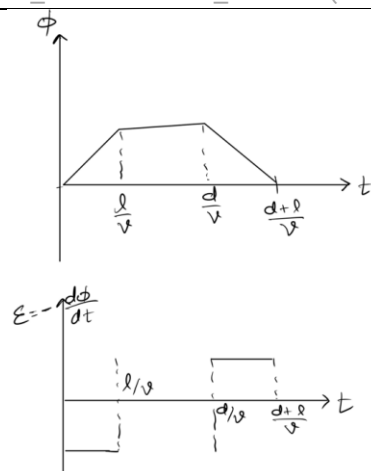
$$= \left[\frac{0.03}{0.60} + 2 \left[\frac{0.01}{50} \right] + \left[\frac{0.05}{10} \right] \right] \times 100$$

$$= \frac{3}{0.6} + \frac{2}{50} + \frac{5}{10}$$

$$= 5 + 0.04 + 0.5 = 5.54 \%$$

Q16 A rectangular loop is moving with constant velocity (v) enters in an uniform magnetic field B, then draw graph of induced EMF versus time.



Sol.

Q17 A spherical body is dropped in glycerin $\left(\rho_{\text{gly}} = \frac{1}{2}\rho_{\text{body}}\right)$. Find viscous force when the body attains terminal velocity.

Sol. $F = 6\pi\eta r v_T$

$$v_T = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

Q18 Given two diatomic gases of adiabatic exponent γ_1 and γ_2 respectively. First gas has also vibrational degree of freedom.

- (1) $\gamma_1 < \gamma_2$
- (2) $\gamma_1 > \gamma_2$
- (3) $\gamma_1 = \gamma_2$
- (4) γ_1 and γ_2 cannot be calculated

Sol. For diatomic gas

$$\frac{C_p}{C_v} = \gamma_1 \text{ has vibrational d.o.f}$$

$$\frac{C_p}{C_v} = \gamma_2 \text{ has no vibration d.o.f}$$

$$\gamma = 1 + \frac{2}{f}$$

$$f = 5, f = 5 + 2(n)$$

$$\int \hat{i} = \frac{2}{f} \downarrow \Rightarrow \gamma \downarrow$$

Q19 A capacitor is connected in a circuit. If the area of plate of capacitor is 16 cm^2 and conduction current is 6 A . Find the displacement current through 3.2 cm^2 area in between capacitor.

Sol. $A = 16 \text{ cm}^2, i = 6 \text{ A}$

$$i_d = \frac{\epsilon_0 d\phi_E}{dt}$$

$$1.2 = \frac{1}{5} \left(\frac{\epsilon_0 d\phi_E}{dt} \right)$$

Q20 An electron having speed v enters in uniform magnetic field B having angular momentum $mvr = \frac{nh}{2\pi}$, by assuming Bohr's postulate holds good. Find the radius of electron in first excited state.

Sol. $mvr = \frac{nh}{2\pi}$, $\frac{mv^2}{r} = qvB$, $n = 2$

$$r = \sqrt{\frac{nhqB}{2\pi}}$$

Q21 A tube containing liquid of mass $2M$ and length $1m$ is rotating about one of its end. If the force exerted by fluid on the other end of tube is F , then find the angular velocity of the tube.

Sol. $F = P \times A$

$$P = \rho\omega^2/2$$